

BUIDL CTC 2026 FALL • AI TRACK

The AI lender that has to **earn its own credit.**

MIRA doesn't put an AI on top of a lending protocol. It makes the AI itself an accountable economic actor — one that can lose authority when it's wrong, and earn it back when it's right.

BUILD: github.com/sodiq-code/mira

LIVE: mira-credit-agent.vercel.app

DEMO: youtu.be/slRPZigX6Xo

THE PROBLEM

AI agents can move money. They can't earn trust.

Today's AI x crypto projects trust the agent by default. Four gaps follow — each one a way an autonomous financial agent can go wrong without consequence.



GAP 01

Unverified inputs

The agent can claim a wallet has a particular history. Off-chain data feeds are trusted, not proven. A compromised worker fabricates the borrower's track record.



GAP 02

Unbounded decisions

The LLM can produce a decision that exceeds the intended risk policy — a larger amount, a lower rate, a longer term. Nothing on-chain enforces the boundary.



GAP 03

No agent accountability

If the AI makes bad loans, there is no persistent on-chain reputation attached to the agent itself. It fails, gets restarted, and tries again — with the same authority it had before.



GAP 04

No cross-chain evidence

What happened on Ethereum stays on Ethereum. Without Attestcoin, the Creditcoin contract cannot verify a real Sepolia repayment — it can only take the worker's word.

THE SOLUTION

A credit system for autonomous financial agents.

The AI gets its own credit score. Good loans raise it; bad loans lower it. The score deterministically gates how much capital the agent is allowed to control.



`score = 500 + repaid × 10 - defaulted × 25 · enforced on-chain in AgentReputation.sol`

ARCHITECTURE

Four layers. One closed loop. No trust gaps.

The AI proposes. Policy constrains. Attestcoin proves. The contract enforces. No single layer trusts the layer above it.

AI

MIRA AI intelligence · risk interpretation

LLM evaluates the verified feature vector and proposes loan terms. Bounded — cannot exceed the policy envelope.

↓ bounded decision

**Creditcoin Policy** governance · deterministic rules

10 on-chain checks: paused, amount, agent tier, borrower tier, rate bounds, term, liquidity, expiry, nonce, evidence hash.

↓ verified evidence

**Attestcoin Protocol** trust root · cryptographic proof

BlockProver precompile verifies Sepolia inclusion proofs on-chain — inside the Loan contract, not by the worker.

↓ source transaction

**Ethereum Sepolia** source of truth

Real borrower activity — stablecoin transfers, wallet age, DeFi positions. The evidence MIRA underwrites against.

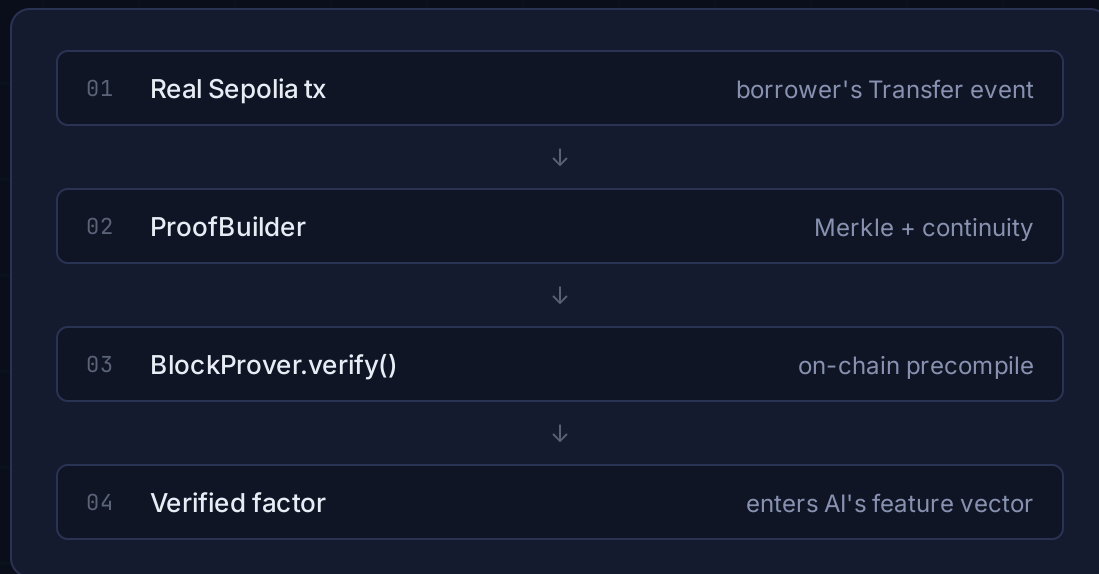
Evidence → Policy → Capital → Outcome → Reputation → Capital authority → next decision

ATTESTCOIN INTEGRATION

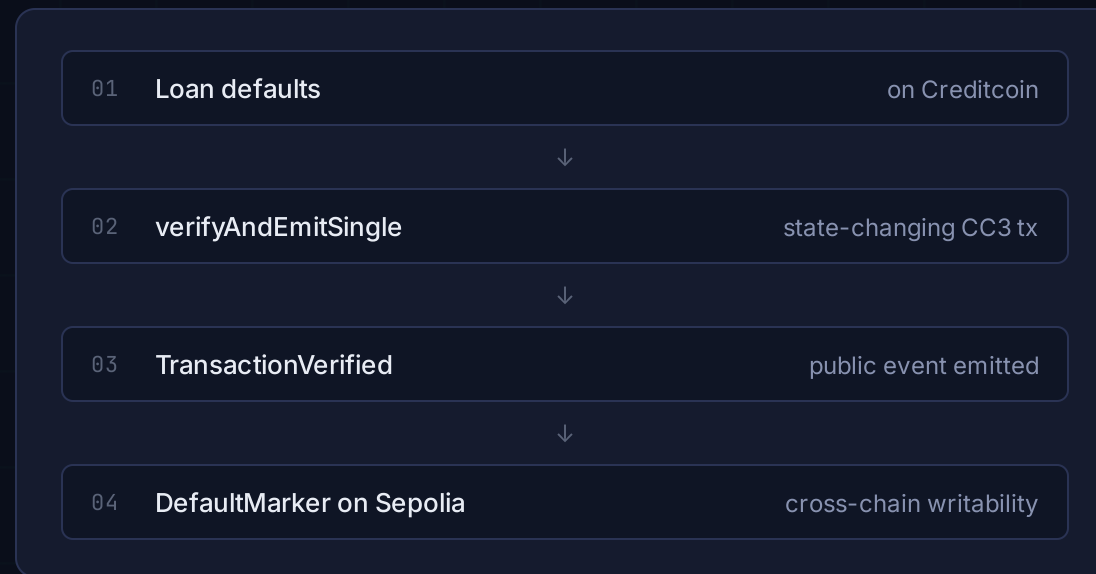
Attestcoin isn't a feature we added. It's the reason MIRA can exist.

MIRA uses both directions of the Attestcoin Protocol — read (Sepolia → Creditcoin) and write (Creditcoin → Sepolia). Most integrations use only one.

READ PATH · SEPOLIA → CREDITCOIN



WRITE PATH · CREDITCOIN → SEPOLIA



CONTRACT-SIDE VERIFICATION – THE CRITICAL INTEGRATION

The Loan contract calls **BlockProver.verify()** on-chain inside **markRepaidWithProof**. The worker cannot fabricate a repayment — the contract requires the cryptographic proof itself.

LIVE PROOF · CC3 TESTNET

Every number is on-chain. Verify it yourself.

Deployed contracts, production-mode locked, real ERC-20 capital, real Sepolia evidence. Not a simulation.

26

LOANS ORIGINATED

22

REPAID

3

DEFAULTED

\$25

CAPITAL AUTHORITY

► PROOF-VERIFIED REPAYMENT TX

0x22593af0d902648480704e9db75e32ff3291f92d56e9ce91a68ebc641f0f127a

Contract called BlockProver.verify() on-chain · moved real tokens · score 675 → 685 (+10)

► PRODUCTION-LOCK TX

0xb38fd9233853e8da872951931eb5d3f6ff7a06db633c4f7da0a313a21369804a

demoMode() = false · worker-trusted markRepaid() path permanently closed

► 6 ADVERSARIAL ATTACKS · ALL DEFENDED ON-CHAIN

- ✗ Malicious LLM (\$10K @ 1%) REJECTED
- ✗ Fabricated proof REJECTED
- ✗ Wrong borrower REJECTED
- ✗ Expired evidence REJECTED
- ✗ Fake repayment REJECTED
- ✗ Insufficient liquidity REJECTED

If autonomous agents are going to control capital, they'll need **credit** — not just wallets.

MIRA is the missing layer: an agent creditworthiness primitive built natively on Creditcoin, verified by Attestcoin, and ready to scale with the ecosystem.

PILLAR 01

Live & Credible Product

26 loans originated, production-mode locked, 41 contract tests passing. Not a prototype — a deployed protocol the Creditcoin team can audit today.

PILLAR 02

Cross-Chain by Design

Uses both directions of Attestcoin — read (Sepolia → Creditcoin) and write (Creditcoin → Sepolia). Deepens the protocol, not just integrates it.

PILLAR 03

Ecosystem Expansion

Agent credit is a new primitive for Creditcoin — the foundation for AgentFi, reputation-weighted DeFi, and undercollateralized lending across the ecosystem.

MIRA could not exist without **Creditcoin** and **Attestcoin**.

github.com/sodiq-code/mira
mira-credit-agent.vercel.app
youtu.be/slRPZigX6Xo